

(iv) Explain whether the work done on the block is positive or negative.

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.....

..... [2]

(v) Use the first law of thermodynamics to determine, to three significant figures, a value for the specific heat capacity of aluminium. Explain your reasoning. Give a unit with your answer.

specific heat capacity = unit [3]

(c) Without further calculation, suggest with a reason how doubling the pressure in (b) is likely to affect the answer in (b)(v).

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..... [1]

[Total: 13]

